

## src/components/CodeMapLoadingPreview.tsx

Kind: added · Baseline: 9aadd5eb5604 · Target: NovaDiff · Generated: 2026-05-26T19:36:27.613Z

### Overview

A new component `CodeMapLoadingPreview` is added at `src/components/CodeMapLoadingPreview.tsx` (lines 1-109). It renders a visual loading state for the code-map feature, consuming a `CodeMapLoadSnapshot` and an optional `exiting` flag.

### Key changes

- **Imports** (R1-R4): `useMemo` from React, `useBackgroundActivity` from `../app/BackgroundActivityContext`, the `CodeMapLoadSnapshot` type, and the local CSS file.
- **Props interface** (R6-R9): `snapshot: CodeMapLoadSnapshot`; `exiting?: boolean`.
- **Component logic** (R18-R50):
  - `useBackgroundActivity` supplies background tasks.
  - `rows` is memoized (`useMemo`) to combine the snapshot phase with other activities, capped at three entries.
  - `primaryPct` is clamped between 2% and 100% from `snapshot.percent`.
- **Render output** (R54-R109): a viewport with stars, horizon, mesh, and a beam whose width reflects `primaryPct`; a console listing up to three status tracks, each showing label, percent, and detail. ARIA attributes ( `role="status"` , `aria-live="polite"` , `aria-busy` ) are present.

### Impact

- **Correctness:** Adds a typed, isolated UI component; no existing code is modified.
- **Maintainability:** Loading logic is encapsulated; future UI changes can be made here.
- **Performance:** `useMemo` limits recomputation; rendering is lightweight.
- **Compatibility:** Requires `BackgroundActivityContext` to be provided; otherwise `useBackgroundActivity` will throw.
- **Observability:** Screen-reader feedback is improved via `aria-live` and `aria-busy`.

### Risks & follow-ups

- **Context availability:** Verify that `BackgroundActivityContext` wraps the component tree where this preview is used.
- **Snapshot shape:** Callers must supply a valid `CodeMapLoadSnapshot` (fields `phase` , `detail` , `percent` ).
- **CSS dependency:** Ensure `CodeMapLoadingPreview.css` exists and defines the referenced class names.
- **Build & lint:** Run `npm run lint` , `npm test` , and `npm run build` to confirm TypeScript and style integration.